

HOMEWORK 8
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 3.1 - CONVERGENT SERIES OF ANALYTIC FUNCTIONS

1. Show that the series $\sum_{n=0}^{\infty} \frac{1}{n^2 + z^2}$ converges on the set $\mathbb{C} \setminus \{z = ni : n \in \mathbb{Z}\}$. Then, show that this convergence is uniform and absolute on each closed disk contained in this region.
2. Show that $\sum_{n=1}^{\infty} \frac{1}{n!z^n}$ is analytic on $\mathbb{C} \setminus \{0\}$. Compute its integral around the unit circle.
3. If $f_n(z) \rightarrow f(z)$ uniformly on a region A , and if f_n is analytic on A , is it true that $f'_n(z) \rightarrow f'(z)$ on A ?